System Category: GAMP 5 Category 4 – Configurable Software

1. Purpose

The purpose of this URS is to define user requirements for the Laboratory Sample Tracking System to ensure it supports laboratory operations, maintains data integrity, and complies with regulatory expectations including 21 CFR Part 11 and ALCOA+ principles.

2. Scope

This URS covers functional and data integrity requirements for the configurable Sample Tracking System used to manage laboratory samples, analyst assignments, results entry, audit trail maintenance, and report generation within the validated Azure environment.

2.1 Out of Scope Vendor =hosted infrastructure, laboratory equipment qualification, and corporate network services beyond the application configuration are outside the scope of this URS

3. Definitions

URS | User Requirements Specification
IQ | Installation Qualification
OQ | Operational Qualification
PQ | Performance Qualification
ALCOA+ | is a framework for ensuring data integrity, emphasizing that data must be Attributable, Legible, Contemporaneous, Original, Accurate, Complete, Consistent, Enduring, and Available
21 CFR Part 11 | is an FDA regulation that sets the standard for electronic records and electronic signatures ensuring they are trustworthy, reliable, and equivalent to paper records and handwritten signatures
GAMP 5 | a set of guidelines developed by International Society for Pharmaceutical Engineering (ISPE). It provides a framework to ensure the quality, safety, and compliance of automated systems in the pharmaceutical and like sciences industries. GAMP 5 emphasizes a risk-based approach to computer system validation, helping professionals manage automated systems throughout their lifecycle

4. System Description

The system will be used to:
1.Track laboratory samples
2. Assign analysts
3. Record results
4. Maintain audit trails
5. Export reports
Hosted in Microsoft Azure.

5. User Roles & Responsibilities

Analyst – Enter results
Reviewer – Review data
QA – Approve & lock records
Admin – Manage user access

6. Functional Requirements

Functional Requirements identify the configurable application functions necessary to support laboratory operations, including samples tracking, role=base access, audit trail maintenance, result entry, and report generation. Each requirement will be verified during IQ/OQ/PQ testing to confirm compliance with GAMP 5 risk-based validation and data integrity principles.

6.1 Security & Access

URS-01 System must require unique username and password login.
URS-02 System must support role-based access control.
URS-03 System must lock account after 5 failed login attempts.

6.2 Audit Trail

URS-04 System must record audit trail for all data changes.
URS-05 Audit trail must capture user ID, timestamp, old value, new value.
URS-06 Audit trail must be reviewable by QA.

6.3 Sample Tracking

URS-07 System must generate unique sample ID.
URS-08 System must store sample metadata (date, analyst, test type).
URS-09 System must allow result entry and review.

6.4 Data Integrity

URS-10 System must prevent deletion of approved records.
URS-11 System must maintain data per ALCOA+ principles.
URS-12 System must require electronic signature for approval.

6.5 Reporting

URS-13 System must generate summary reports.
URS-14 System must export data to Excel.
URS-15 Reports must include audit trail references.

6.6 Backup & Recovery

URS-16 System must perform daily automated backups.
URS-17 System must allow data restoration.

7. Non-Functional Requirements

URS-18 System must support 20 concurrent users.
URS-19 System uptime must be ≥ 99%.
URS-20 System response time ≤ 3 seconds.

8. Acceptance Criteria

Each requirement will be verified during IQ/OQ/PQ testing and documented in the Traceability Matrix.

9. Document Approval

Validation Lead – Brian Alston
QA Reviewer – TBD
System Owner – TBD